

Two Wright–Fisher updates and their stationary marginals

Setup

Each individual carries a mutation count $k \in \mathbb{N}_0$. We use geometric selection $w_k = (1 - s)^k$ ($0 < s < 1$) and Poisson mutation at rate $u \geq 0$ (per individual, per generation), so the new mutation count of an offspring is the parent’s class plus an independent $\text{Poi}(u)$.

Let Q denote a probability distribution on $\mathbb{N}_0$ (the marginal class distribution of one individual). Two basic operators act on such Q :

$$(M_u Q)(k) = (Q * \text{Poi}(u))(k) = \sum_{j=0}^k Q(j) \text{Poi}(u; k - j) \quad (\text{mutation}),$$

$$(S Q)(k) = \frac{w_k Q(k)}{\sum_{j \geq 0} w_j Q(j)} \quad (\text{selection / reweighting by } w).$$

We compare two update rules per generation:

$$(A) \text{ “sel} \rightarrow \text{mut” : } T_A := M_\mu \circ S, \quad (B) \text{ “mut} \rightarrow \text{sel” : } T_B := S \circ M_u.$$

Multinomial resampling of N offspring is applied afterwards; for the analysis of the marginal we look at the deterministic large- N limit and return to fluctuations at the end.

Two Poisson identities

The reason the calculation is short is that both operators preserve the Poisson family:

$$S(\text{Poi}(\nu)) = \text{Poi}(\nu(1 - s)), \quad (1)$$

$$M_u(\text{Poi}(\nu)) = \text{Poi}(\nu + u). \quad (2)$$

(Eq. (1) follows from $w_k \text{Poi}(\nu; k) = (1 - s)^k e^{-\nu} \nu^k / k! \propto (\nu(1 - s))^k / k!$, i.e. $w_k = (1 - s)^k$ is an exponential tilt of Poisson and merely rescales its parameter by $1 - s$ (equivalently, exponential selection with $s' = -\ln(1 - s)$). Eq. (2) is the standard sum-of-independent-Poissons identity.)

Stationary Poisson marginals

Suppose the marginal is $Q = \text{Poi}(\theta_{\text{eq}})$ and we look for fixed points of T_A, T_B .

Model A (sel $\rightarrow$ mut). By (1) and (2),

$$T_A(\text{Poi}(\theta_{\text{eq}})) = M_\mu(\text{Poi}(\theta_{\text{eq}}(1 - s))) = \text{Poi}(\theta_{\text{eq}}(1 - s) + \mu).$$

The fixed-point equation $\theta_{\text{eq}} = \theta_{\text{eq}}(1 - s) + \mu$ gives

$$\boxed{\theta_{\text{eq}}^{(A)} = \frac{\mu}{s}}. \quad (3)$$

Model B (mut $\rightarrow$ sel). Now

$$T_B(\text{Poi}(\theta_{\text{eq}})) = S(\text{Poi}(\theta_{\text{eq}} + u)) = \text{Poi}((\theta_{\text{eq}} + u)(1 - s)),$$

so $\theta_{\text{eq}} = (\theta_{\text{eq}} + u)(1 - s)$ yields $\theta_{\text{eq}} s = u(1 - s)$ and hence

$$\boxed{\theta_{\text{eq}}^{(B)} = \frac{u(1 - s)}{s}}. \quad (4)$$

For small s , both formulas reduce to $\theta_{\text{eq}} \approx u/s$, but for finite s they differ; in particular

$$\frac{\theta_{\text{eq}}^{(B)}}{\theta_{\text{eq}}^{(A)}} = \frac{u}{\mu} \cdot (1 - s).$$